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标题	Highly Enantioselective Synthesis of 3,3-Diarylpropyl Amines and 4-Aryl Tetrahydroquinolines via Ir-Catalyzed Asymmetric Hydrogenation
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摘要（原文节选）

Chiral nitrogen-containing compounds are crucial for the chemical, pharmaceutical, and agrochemical industries. Nevertheless, the synthesis of certain valuable scaffolds remains underdeveloped due to the vast chemical space available. In this work, we present a diastereoselective methodology for synthesizing 3,3-diarylallyl phthalimides, which, following iridium-catalyzed asymmetric hydrogenation using Ir – UbaPHOX, yield 3,3-diarylpropyl amines with high enantioselectivity (98 – 99% ee). The importance of alkene purit ⋯

数据表 · Table 1（11 行，含数值行 11 行）

Optimization of Pressure, Solvent, and Catalyst Loading Parametersa

entry	catalyst loading [mol %]	PH2 [bar]	solvent	conv. [%]b	ee [%]c
1	5	50	DCM	>99	96
2	5	50	TFT	>99	97
3	5	50	DCE	>99	96
4	5	50	toluene	62	95
5	5	50	EtOAc	3	–
6	5	50	DMC	0	–
7	5	50	PC	0	–
8	5	10	DCM	>99	91
9	5	3	DCM	53	57
10	1	50	DCM	>99	96
11d	1	50	DCM	>99	98

脚注（逐字）：aThe experiments were carried out at 0.17 M.bDetermined by 1H NMR analysis of the crude reaction mixtures.cDetermined by HPLC analysis on a chiral stationary phase.dThe starting material was synthesized via Route B (Scheme 3b).

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